

SWAPNIL MITRA

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SUMMARY

Software Engineer with 3+ years of experience in system architecture, enterprise middleware integration, and AI-enabled tooling. Proficient in .NET Core, Azure (Functions, API Management, Logic Apps), RabbitMQ, React, and cloud-native distributed systems. Seeking software engineering roles building scalable, high-availability production services.

EDUCATION

Indian Institute of Technology, Kharagpur

Graduated May 2023

Bachelor of Technology, Biotechnology and Biochemical Engineering

SKILLS

Languages: JavaScript, TypeScript, C#, Python, Rust, SQL, PostgreSQL, Solidity

Frameworks / Libraries: .NET Core, ASP.NET, React, Next.js, Flutter, Express, Fluent UI, MSAL, LangChain, Streamlit, Tailwind CSS, Wagmi

Cloud & Middleware: Azure Functions, Azure API Management, Azure Logic Apps, Azure DevOps, GitHub Actions, RabbitMQ, Docker, AWS (S3, Bedrock, Transcribe, Polly)

Tools & Platforms: Prometheus, Grafana, Microsoft Graph API, SharePoint, SQL Server, Git, Hardhat, Prisma, Drizzle, pgvector, Pinata (IPFS), Vitest

EXPERIENCE

Research and Innovation Engineer

Jul 2023 – Present

Adrosonic, Mumbai

- Leading a **team of 4 engineers** as **Technical Lead** on an enterprise **broker management platform**, owning **solution architecture** across four delivered workstreams — **SSO** for the broker-agency user base, then **quote-bind**, **cancellation**, and **amendment** insurance flows — on **Azure cloud-native** infrastructure spanning serverless compute, API management, messaging, and CI/CD.
- Architected the **middleware integration layer** for that same platform, carrying its quote-bind, cancellation, and amendment flows out to **downstream quote-aggregator services in Australia**; built a **JSON transformation engine via Azure Functions**, configured **Azure API Management** with OAuth 2.0 (external) and Managed Identity (internal) gateways with custom policies, and instrumented telemetry via **Azure Application Analytics** — reducing integration development time from **~6 weeks to ~2 weeks** with a **99.9% successful payload delivery rate**.
- Re-engineered an **email filing agent and DMS pipeline** (SharePoint) to **.NET Core + RabbitMQ** pub/sub; implemented **message deduplication**, dead-letter queuing, and **queue mirroring across RabbitMQ cluster nodes** containerised with **Docker**; built **Prometheus**-instrumented observability dashboards — achieving **~40% reduction in average message latency** and improved throughput under peak load.
- Migrated a legacy COM-based Outlook add-in to a modern **React + Fluent UI Outlook Web Add-in** with **SSO via MSAL (Microsoft Authentication Library)**, Microsoft Graph API integration, and offline attachment caching via IndexedDB; deployed to **10,000+ users** with measurably reduced UX friction.

PROJECTS

MemVault 🌐 github.com/swap-mitra/memvault

Aug 2026

Local-first, hash-chained memory engine for AI agents with fully explainable retrieval.

- Built a local-first memory engine for AI agents in **Rust** that records every fact and every retrieval in an **append-only, hash-chained ledger**; cryptographic erase destroys the content key while the chain still verifies end to end, so deletion is provable rather than trusted.
- Implemented **hybrid retrieval** fusing **usearch ANN** and **BM25** ranks by **reciprocal rank fusion** with decay weighting and token-budget cutoffs, emitting a per-candidate provenance table that replays any past retrieval from its id; shipped the engine as a CLI, an MCP stdio server (optional gRPC), and **PyO3** in-process Python bindings.

Groundwork 🌐 github.com/swap-mitra/groundwork

2024 – 2026

Evidence-grounded outbound email drafting gated behind explicit human approval.

- Built a multi-tenant outreach workspace (**Next.js**, **Drizzle**, **Postgres + pgvector**) that ingests a job URL behind an **SSRF guard** with a headless-browser fallback, then drafts evidence-cited email via **OpenRouter** using **hybrid retrieval** (lexical + vector + RRF + MMR) over a **halfvec HNSW** index.
- Enforced **human-in-the-loop** delivery with append-only draft versions, reviewer-attributed approvals, and a full activity audit trail; ran ingestion and generation as **Inngest** workflows with bounded retries, durable per-workspace Postgres rate limits, and Gmail draft-only export behind a swappable provider interface.

City Vault 🌐 github.com/swap-mitra/city-vault

2025 – 2026

Multi-tenant records governance with 6-role RBAC, full document lifecycle, and IPFS audit trail.

- Architected a multi-tenant records governance platform (Next.js 15, Prisma, Postgres in Neon DB) featuring 6-role RBAC, full document lifecycle management, IPFS pinning via Pinata for content-addressed storage, and append-only audit trails.
- Implemented a sliding window rate limiter with per-key tracking, built a comprehensive Vitest suite covering route handlers and governance workflows, and designed a brutalist interface using Tailwind CSS 4.

AGRISETU — AI Voice-First Agricultural Marketplace 🌐 github.com/hackerslash/agrisetu

Feb – Apr 2026

Voice-first marketplace for farmers to procure supplies via collective group ordering.

- Engineered a voice-first agricultural marketplace (Flutter, Next.js, Express, Turborepo) processing PCM audio through AWS Transcribe (11 languages), GLM-4.7 Flash on Bedrock, and Polly TTS, with automated deployments via GitHub Actions.
- Designed and deployed a RAG pipeline utilizing Titan Text Embeddings v2 and Bedrock Knowledge Base to retrieve semantic domain knowledge, eliminating LLM hallucinations on regional dialect queries.

EXTRA-CURRICULARS

- National Service Scheme Volunteer — Awarded Silver Medal for collaborative performance (2019).
- Associate Manager, Entrepreneurship Cell, IIT KGP — Organised the Global Entrepreneurship Summit 2021, LSM 2020, EAD 2020, and Empresario 2020.
- Club Service Director, Rotaract Club of Calcutta North East.